

AYMANE AIT DADS

AI Engineering Intern | Edge AI Inference · LLM Fine-Tuning · Multimodal Systems

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by fine-tuning a 30B-parameter Mamba-Transformer with LoRA SFT on structured reasoning tasks requiring fundamentally different strategies per category. Reduced manual network analysis time by 60% at Orange Maroc by building automated ML pipelines on 100K+ records, and ranked 1st in class on an AI-generated image detection benchmark (0.791 AUC) with a CLIP ViT-L/14 backbone submitted to NTIRE 2026 at CVPR. Brings hands-on PyTorch expertise, ablation-driven experimentation methodology, and experience optimizing inference pipelines with BF16 mixed precision and vLLM - directly applicable to building low-latency Voice-to-Action command loops on edge AI hardware. Aims to apply multimodal fine-tuning and rapid prototyping skills at Destinus to advance autonomous drone flight intelligence for mission-critical defense applications.

CORE COMPETENCIES

Edge AI Hardware | Low-Latency AI Pipelines | Vision Language Models | Multimodal Fine-Tuning | Rapid Prototyping | Inference Optimization | Autonomous Systems | Embedded AI Systems

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024
Data Science Intern - Casablanca, Morocco

- Built an automated machine learning pipeline combining Random Forest and K-Means on **100K+ fiber-optic network records**, classifying quality across thousands of nodes for real-time network optimization.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Pandas-based data preprocessing at scale.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, bridging **ML model results** to non-technical operational decisions using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge Kaggle · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, guiding targeted data curation of 605 verified cipher rows and 64 verified 8-bit operation rows.

PyTorch · Unsloth · TRL · PEFT · vLLM · Hugging Face Transformers · Triton

AI-Generated Image Detection EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M parameters) with a custom MLP head trained on 250K images spanning 25+ generator types; separately submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC)** for out-of-distribution generalization; applied 10-view TTA (flips, crops, blur, resizes) and a 3-model ensemble weighted by validation AUC under FP16 mixed precision on a T4 GPU.

PyTorch · OpenCLIP · ViT-L/14 · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM Sep 2023 - Present
Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Image Security, Cloud Computing, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

LLM Fine-Tuning & Inference: LoRA · QLoRA · SFTTrainer · vLLM · Unsloth · TRL · PEFT · BF16/FP16 Mixed Precision

ML Frameworks & Computer Vision: PyTorch · TensorFlow · Scikit-learn · OpenCLIP · ViT · CNN · DenseNet · Hugging Face Transformers

MLOps & Infrastructure: Python · Linux · Git · Docker · CUDA · Triton · Kaggle GPU · Ablation Studies · REST APIs